

IIT TIRUPATI NAVAVISHKAR I-HUB FOUNDATION

Building India's Future in Positioning & Precision Technologies

A website design report for IITTNiF — the national Technology Innovation Hub under NM-ICPS, anchored at IIT Tirupati.

Prepared as a first-principles blueprint for identity, architecture, and engineering.

RESEARCH

MANDATE

ARCHITECTURE

DELIVERY

NATIONAL IMPACT

CONTENTS

Map of the Report

Twelve sections, each carrying one stage of our Research → National Impact colour signature, explained in full in Section 03.

	01	EXECUTIVE SUMMARY A Platform, Not a Website	P. 03
	02	UNDERSTANDING THE MANDATE Who This Is Actually For	P. 04
	03	DESIGN PHILOSOPHY Deriving Color From First Principles	P. 05
	04	INFORMATION ARCHITECTURE The Eight Tabs, One Spine	P. 06
	05	PAGE-BY-PAGE BLUEPRINT Anatomy of Each Page	P. 07
	06	WEBSITE FLOW Five Journeys Through One Ecosystem	P. 08
	07	TECHNICAL ARCHITECTURE Built for a 20-Year Institution	P. 09
	08	COMPONENT SYSTEM One Skeleton, Eight Expressions	P. 10
	09	CONTENT & CMS MODEL Who Edits What, and How	P. 11
	10	THEMING RULES Light by Default, Dark by Exception	P. 12
	11	DELIVERY PLAN Three Weeks, Three Phases	P. 13
	12	CLOSING Why NS OJAS Media Labs	P. 14

Our Approach to This Site

We're not designing a college department's homepage. Our starting judgment: IITTNiF is a national Technology Innovation Hub under NM-ICPS, so we're treating its website as the front door of a country-scale mission in Positioning & Precision Technologies — and designing accordingly.

WHAT WE'RE AVOIDING

The brief asks us to stay away from generic templates, Joomla holdovers, and event-listing pages that flatten a deep-tech mission into a directory. We agree, and we've designed against that risk directly.

WHAT WE NOTICED

Every page in the brief routes back to one platform — VIKAS. We've made that the job of our architecture: make the spine visible, not eight disconnected tabs.

WHAT WE'RE PROPOSING

A first-principles design system — colour, type, and component logic all derived from what IITTNiF does, not picked from a mood board. Section 03 shows our working.

WHAT THIS REPORT COVERS

Five things, in order: who we're designing for (Sec. 02); how we derived the visual identity rather than chose it (Sec. 03); how we're treating eight tabs as one connected ecosystem (Sec. 04–06); how we intend to build and hand it over (Sec. 07–10); and our delivery path (Sec. 11–12).

8

NAVIGATION TABS

1

CONNECTIVE SPINE — VIKAS

3

WEEK BUILD RUNWAY

THREE THINGS WE'RE ADDING BEYOND THE BRIEF

Bilingual layer. English/Hindi toggle by default — a national mission's front door shouldn't have a language gate.

An open Impact feed. Counters and outcomes exposed as a simple API, so partner government dashboards can embed live IITTNiF data — extending the Digital Governance work IITTNiF already does into the site itself.

Inline glossary. Hover-tooltips on terms like GNSS Reflectometry or GeoAI, so a visiting ministry official isn't lost three sentences into the Technology page.

02 UNDERSTANDING THE MANDATE

Who We're Designing For

We're going in knowing IITTNiF carries real institutional weight: a Section 8, not-for-profit Technology Innovation Hub under NM-ICPS, Department of Science & Technology, Government of India — anchored at IIT Tirupati. Our design has to hold that weight while still feeling like the front end of a living, technology-building ecosystem, not an extension of a government circular.

SEVEN STAKEHOLDERS, SEVEN DIFFERENT FIRST QUESTIONS — OUR READ

Startup

Wants proof IITTNiF can get a prototype to pilot — not another incubator pitch deck.

Industry

Wants a credible technology partner for validation, licensing, and deployment at scale.

Government

Wants a mission-grade partner for digital governance, defence, and national infrastructure.

Faculty / Researcher

Wants lab access, funded TDPs, and a route from research to real deployment.

Student

Wants a visible, concrete path — fellowship, hackathon, internship — into the ecosystem.

Mentor / Expert

Wants to see the calibre of problems and people before committing time.

International Partner

Wants evidence of India-scale credibility and a clear collaboration entry point.

WHAT WE'RE DESIGNING AROUND

No single visitor wants the whole site, so we're making the Home page's job triage, not storytelling. Every major section has to hand the visitor to their specific pathway within the first scroll or two — primarily through VIKAS.

AN EIGHTH NEED WE NOTICED, NOT IN THE BRIEF

A startup or researcher who already applied through VIKAS doesn't want to re-read the site — they want to know where their application stands. We're proposing a lightweight, login-free status check (a magic link tied to their submission) so this doesn't become a daily flood of "any update?" emails for TIH staff.

Deriving Color From First Principles

We didn't choose colour from a mood board — we derived it, in four steps: read the semantic content already inside IITTNiF's own brief, map that meaning onto hue and temperature, apply formal contrast and accessibility rules, then stress-test the result against every content type on the site.



RESEARCH

#10182B

Deep Indigo — institutional trust, the night sky a satellite is tracked against.



MANDATE

#1C5C8C

Signal Blue — the cool, instrument-panel blue of positioning and navigation systems.



ARCHITECTURE

#0E8C82

Geospatial Teal — the data-and-validation colour of GIS and earth-observation tooling.



DELIVERY

#D98A2B

Beacon Amber — the warm, precision-instrument amber of active deployment.



NATIONAL IMPACT

#C1502E

Earth Terracotta — tangible, ground-level, the colour of deployed outcomes.

OUR SIGNATURE: A CHROMATIC SPINE

IITTNiF's brief repeats one diagram on nearly every page: Research → Technology Development → Prototype → Validation → Deployment → National Impact. Rather than inventing a separate visual language, we're running that exact pipeline — in this report and on the website itself — as a colour temperature gradient: cool and abstract at the conceptual end, warm and tangible at the deployed end. Every section of this document, and every major page we'll build, sits at its true position on that single gradient. That's our central design decision: colour does information work here, it doesn't just decorate.

TYPE SYSTEM — THREE OPEN-LICENSE FACES, EACH EARNING ITS ROLE

Aa

Space Grotesk

Display & headings.

Geometric, instrument-grade — reads like precision hardware, not a startup deck.

Aa

Public Sans

Body text. The U.S. federal government's own public-domain typeface — fitting pedigree for a DST-anchored mission.

Aa

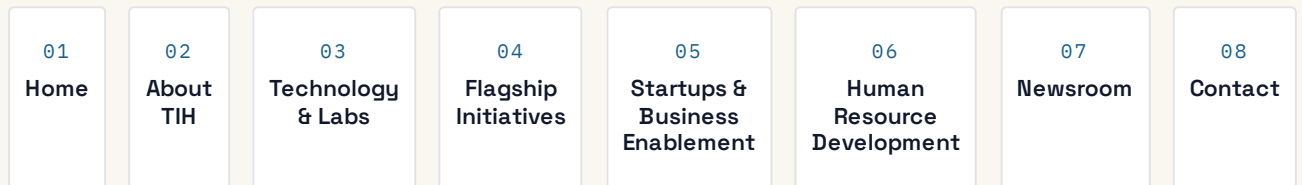
IBM Plex Mono

Data, counters, labels, captions. Engineering-instrument character for every number on the site.

04 INFORMATION ARCHITECTURE

The Eight Tabs, One Spine

Our reading of the brief: eight tabs on the surface, but underneath, every one of them is a doorway into the same engagement platform — VIKAS. We're keeping the navigation flat and simple; the architecture behind it is what keeps funnelling every stakeholder back to one place.

**VIKAS — SINGLE-WINDOW ENGAGEMENT PLATFORM**

Innovation calls · startup engagement · mentor network · government challenges · industry problem statements

TAB-BY-TAB PURPOSE, IN ONE LINE EACH

Home	20-second proof of identity, then triage every visitor to their pathway.
About TIH	Identity, mandate, governance, leadership — not a dumping ground for activities.
Technology & Labs	Proof IITTNiF is a technology engine, built around three major laboratories.
Flagship Initiatives	Long-term identity platforms — VIKAS, SPIN Labs, VidyaGIS, Bharath PNT.
Startups & Enablement	Deep-tech venture support framed as deployment, not routine incubation.
HRD	National talent-development mission — not a training calendar.
Newsroom	CMS-driven credibility — news, awards, press, opportunities, kept out of main nav clutter.
Contact	Stakeholder-routed enquiry pathways, not a static address block.

Anatomy of Each Page

When we stripped the content away from Home, Technology, Startups, HRD, and Ecosystem, the same skeleton appeared underneath every one of them. We're treating that repetition as the component system (detailed fully in Section 08), not as monotony. Here's that skeleton, instantiated for three representative pages:



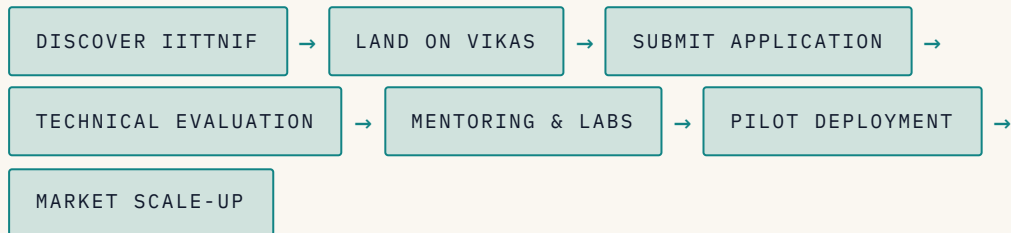
PAGE	HERO	PILLARS	PIPELINE	PORTFOLIO	STORIES	CTA
Home	Cinematic PPT hero	3 Labs as cards	Research → Impact pipeline	4 flagship platforms	Newsroom preview	Choose-your-pathway cards
Technology & Labs	Tech mandate hero	3 labs, deep detail	Innovation pipeline (TRL-aware)	Filterable TDP portfolio	Tech outcomes & counters	Submit a Technology Proposal
Startups	Founder-facing hero	VIKAS Advantage block	Idea → Enterprise journey	Filterable startup cards	Founder success stories	Apply as a Startup

We're carrying the identical pattern into Flagship Initiatives, HRD, and the Ecosystem page, substituting VIKAS, SPIN Labs, learning pathways, and platform cards into Pillars and Portfolio respectively — see the full mapping in Section 08.

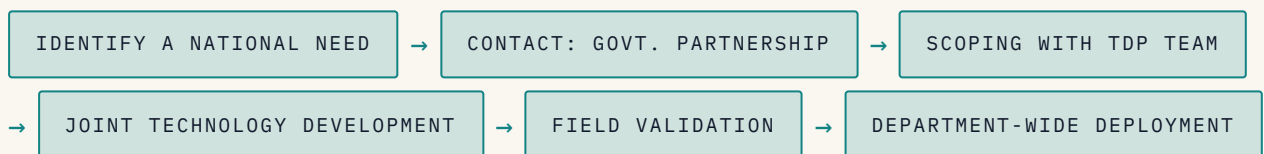
Five Journeys Through One Ecosystem

We test our information architecture by walking it as a visitor would, not by reading it as a sitemap. Two of the five journeys we mapped, in detail:

JOURNEY – "I AM A STARTUP"



JOURNEY – "I AM FROM GOVERNMENT"



THE REMAINING THREE JOURNEYS, MAPPED IN OUR FULL IA WORKBOOK

Faculty / Researcher

Lab access → TDP proposal
→ funded research →
publication / transfer.

Student

Discover → VidyaGIS / SPIN
Labs → challenge or
fellowship → mentorship.

International Partner

Credibility check →
International Partnership
form → MoU pathway.

Built for a 20-Year Institution

We're building this to outlive whoever maintains it next — fast, accessible, SEO-credible, and editable by non-technical TIH staff, without ever drifting into the generic template the brief asked us to avoid.

LAYER	DECISION	WHY
Framework	Next.js — App Router	Server Components keep JS minimal; ISR updates news/opportunities without a full rebuild; native fit for the heavy interactivity the brief demands (filters, counters, maps).
Animation	Framer Motion	Orchestrated reveals and hover micro-interactions for pipeline diagrams, counters, and cards — used deliberately, never decoratively.
Maps	Mapbox / Leaflet	Interactive India network map for SPIN Labs, partners, and collaboration footprint.
CMS	Decision point — see below	Must let TIH staff edit news, events, opportunities, and the startup portfolio without code.
Hosting	Decision point — see below	Domain is already owned by IIT Tirupati; runtime/server choice still shapes how ISR behaves.

CMS — FOR IITTNIIF TO CONFIRM

Given the DST / NM-ICPS lineage, data sovereignty may matter: **Payload** or **Strapi** (self-hosted, full control) are the safer default over a cloud-only SaaS CMS such as Sanity, unless IITTNIIF confirms cloud hosting is acceptable.

HOSTING — FOR IITTNIIF TO CONFIRM

Vercel is the easiest Next.js host but is a foreign commercial cloud. If IITTNIIF's own infrastructure is preferred, we deploy via Docker on their servers — this changes how ISR caching is configured.

CONFIRMED REQUIREMENT — GIGW 3.0 & WCAG 2.1 AA

Given IITTNIIF's DST / NM-ICPS lineage, we're treating **GIGW 3.0 (Guidelines for Indian Government Websites & Apps)** as a confirmed requirement, not an option — including its WCAG 2.1 Level AA accessibility baseline and its CERT-In cybersecurity chapter. We're building to the GIGW mandatory ("MUST") checkpoints from day one, so STQC / Certified Quality Website certification stays realistic later, rather than becoming a costly retrofit.

PERFORMANCE & ACCESSIBILITY BUDGET

Lighthouse Performance	≥ 90 on mobile, all primary pages
Accessibility	WCAG 2.1 AA minimum — the GIGW 3.0 baseline — across all text/background pairs
Compliance	GIGW 3.0 mandatory checkpoints + CERT-In cybersecurity chapter, built-in from the start
Responsive targets	360px → 1920px, tested at 5 breakpoints

One Skeleton, Eight Expressions

We're designing six reusable modules, built once and re-themed per page. This is what makes a 3-week build realistic without producing eight one-off pages — and what keeps the site maintainable long after we hand it over.

MODULE	BEHAVIOUR	WHERE IT APPEARS
Hero	Full-bleed statement + 2 CTAs	Home (cinematic PPT loop) · Startups (founder-facing) · Tech & Labs (global-tech-co tone)
Pillars	3–6 card grid, hover reveal	Home (3 major labs) · HRD (6 learner pathways) · Ecosystem (platform cards)
Pipeline	Animated horizontal flow	Every major page (Research → Impact, re-scoped per page)
Portfolio	Filterable card grid	Tech & Labs (TDP portfolio) · Startups (startup portfolio)
Stories	Photo + narrative + outcome	Startups, HRD, Ecosystem (real journeys, not stats)
Engage / CTA	Stakeholder card router	Home, Ecosystem, Contact (routes into VIKAS)

We're inheriting each module's accent colour from the page's position on our chromatic spine (Section 03) — so one component library produces eight visually distinct pages without eight separate design files.

Who Edits What, and How

We're taking the brief's requirement at face value: TIH staff need to update the site without a developer. That's shaped every recurring content type below into a structured CMS model — never a page builder, and never raw code access.

CONTENT TYPE	OWNED BY	UPDATE CADENCE	NOTES
News & Updates	TIH Comms team	Weekly	Category + date filters, auto-archives
Events (upcoming/past)	TIH Comms team	Ongoing	Calendar + list view, registration links
Startup Portfolio	Startup/VIKAS team	Monthly	Logo, domain, stage, achievement fields
Opportunities (calls, fellowships)	HRD + VIKAS team	As issued	Deadline-aware, auto-expires
Leadership & Lab contact cards	Admin team	Rare	Photo, designation, official email only
Publications & Reports	Comms team	Quarterly	PDF upload + cover image
Page copy (Hero, About text)	Developer-assisted	Rare	Structured fields, not free-form code

A HARD RULE WE'RE ENFORCING AT THE SCHEMA LEVEL

No internal financial, approval, MoU, mobile number, or disbursement detail will ever be exposable through a public-facing CMS field — we're building that restriction into the schema itself, not leaving it to editorial discipline.

Light by Default, Dark by Exception

We're running the site on the light, colour-derived palette from Section 03 as its default and only system. Dark mode isn't a parallel design in our plan — it's a narrow, justified exception.

THE EXCEPTION CLAUSE WE'RE PROPOSING

We'll only use a dark treatment for a section if **all** of the following hold:

1. The content is data-visualisation or satellite/space imagery where dark contrast genuinely improves legibility — not aesthetic preference.
2. It's scoped to that single section only, never inherited by surrounding pages.
3. It still passes the same WCAG AA contrast budget as our light system.
4. We sign it off case-by-case during build, never assume it in advance.

WHERE WE EXPECT THIS TO MATTER MOST

The Technology & Innovation page's own design notes mention "dark and light themes." Under our clause, that becomes: light by default, with a possible dark-mode panel reserved for a satellite/data-visualisation moment inside the Technology Domains or Infrastructure section — never the page shell itself.

Three Weeks, Three Phases

Our plan against the three-week window expected once this PDR is approved — sequential, not overlapping, so each phase de-risks the next.

WEEK 1 Foundation IA sign-off · palette & type lock · content model · wireframes for all 8 tabs · CMS/hosting decisions confirmed with IITTNiF	WEEK 2 Build Component library (Hero, Pillars, Pipeline, Portfolio, Stories, CTA) · all 8 pages assembled · India map & pipeline animations	WEEK 3 Harden & Ship CMS wiring · accessibility & performance pass · responsive QA across breakpoints · staff handoff training · launch
---	--	--

DEPENDENCIES OUTSIDE OUR CONTROL

Final logo & brand assets, CMS/hosting confirmation, and leadership photography/copy sign-off are the three items we'll chase from IITTNiF in Week 1 — they're the only things that can compress or stretch this timeline.

Why NS OJAS Media Labs

This report was built the same way we intend to build the website itself — derived, not decorated. Every colour, every component, every page traces back to what IITTNiF actually is: a national engine for Positioning & Precision Technologies. We don't believe a mission like this deserves a templated front door, and we've shown our reasoning at every step so IITTNiF can hold us to it.